FACULTY OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

[Project Title Goes Here]

FINAL YEAR PROJECT – I (FYP-I)

PROJECT PROPOSAL

(Research-Based Project)

A project proposal submitted in partial fulfilment of the requirements for the degree of

Bachelor of Science in Computer Science

Submitted by: **[Student Name – Registration No.]**
[Student Name – Registration No.]
[Student Name – Registration No.]

Supervised by: **[Supervisor Name, Designation]**

Department of Computer Sciences
Faculty of Computer Science & Information Technology
The University of Lahore, Sargodha Campus

Session: **[20XX – 20XX]** **[Month, Year]**

Project Registration Form

Project ID (for office use only):	
Type of Project	☐ Traditional ☐ Industrial ☐ Continuing
Nature of Project	☐ Development ■ **Research** ☐ R&D
Area of Specialisation	**[e.g. Machine Learning, Federated Learning, Cybersecurity, IoT, NLP]**
Department	Computer Science
Registration No. of FYP-I	**[Reg. No.]** **Date of Registration: [DD-Mon-YYYY]**
Name of Supervisor: [Supervisor Name]	**Co-Supervisor (if any): [Name / None]**
Designation: [Designation]	**Designation: [Designation]**

Tick Research *under Nature of Project for a research-based FYP-I – this indicates the proposal is centred on investigating a problem, hypothesis, or research question rather than solely building a deliverable system.*

Project Group Members

Sr.#	Reg.#	Student Name	CGPA	Email ID	Phone #
(i) Group Leader	**[Reg.#]**	**[Name]**	**[]**	**[Email]**	**[Phone]**
(ii)	**[Reg.#]**	**[Name]**	**[]**	**[Email]**	**[Phone]**
(iii)	**[Reg.#]**	**[Name]**	**[]**	**[Email]**	**[Phone]**

Declaration: FYP group members have cleared all prerequisite courses for FYP-I as per their degree requirements, including:
Object Oriented Programming, Software Engineering, Database Systems-I, Report Writing Skills, and any research-methods course specified by the department.

Signature(s) of Student(s): ____

Summary of the Proposed Research

Write a single, self-contained paragraph of 200–300 words covering: the problem context, the research gap, what you propose to investigate, the method in one sentence, and the expected outcome/contribution. This is often read on its own by the FYP committee, so it should make sense without the rest of the document.

[Type the 200–300 word summary of your proposed research here. Cover: (1) the problem and why it matters, (2) the gap in existing work, (3) your proposed research question/approach, (4) the method you will use, and (5) the expected contribution or outcome.]

Plagiarism-Free Certificate

This is to certify that I, **[Group Leader Name]**, S/D/o **[Father's Name]**, Group Leader of the FYP-I proposal registered under Registration No. **[Reg. #]** at the Department of Computer Science, The University of Lahore, Sargodha Campus, declare that this proposal has been checked by my supervisor and that the similarity index is ________% (an acceptable limit per HEC guidance is generally below 19–20%). The plagiarism/similarity report is attached herewith as Appendix A.

Date: ______________ Name of Group Leader: ______________________ Signature: __________

Run your proposal through the plagiarism tool your department uses (e.g. Turnitin) and insert the similarity percentage before submission – this certificate is not valid without it.

Recommendation by the Research Supervisor

I have reviewed this FYP-I research proposal and recommend it for approval by the FYP Committee.

Name: ____________________________ Signature: ______________________ Date: _________________

Signed by the FYP Supervisory / Evaluation Committee

S.#	Name of Committee Member	Designation	Signature & Date
1	**[Name]**	**[Designation]**	
2	**[Name]**	**[Designation]**	
3	**[Name]**	**[Designation]**	

Approval by the Departmental FYP Committee

Certified that this FYP-I proposal has been reviewed by the members of the Departmental FYP Committee and considered suitable for registration.

Secretary, Departmental FYP Committee Name: ____________________________________ Signature: ____________________________________ Date: ____________________________________
Chairperson / Head of Department, Computer Science Name: ____________________________________ Signature: ____________________________________ Date: ____________________________________

Contents

1 Introduction & Background

Introduce the domain/field, define key terms a non-specialist reader needs, and narrow down from the broad field to your specific problem area. End this section by stating, in one or two sentences, what this proposal will investigate. Typical length: 400–600 words.

[Write your introduction and background here.]

2 Problem Statement

State the problem precisely, in 1–2 short paragraphs. A strong problem statement names the gap or limitation in current systems/approaches, and makes clear why it is worth solving. Avoid restating the whole introduction here.

[Write a precise problem statement here.]

3 Related Work / Literature Review

This section is ***mandatory****. Review previous scientific and development work on your topic; give the historical background and current state of the art. Use only credible sources (journals, conferences, books) and cite every source in the text using IEEE style, e.g. "[1]". A comparison table summarising related work (approach, strengths, limitations) is strongly recommended for research-based FYPs.*

[Summarise 8–15 relevant papers/systems, organised by theme or chronology.]

Table 1: Comparison of related work **[(edit caption)]**

Ref.	**Approach / Technique**	**Strengths**	**Limitations / Gap**
[[1]]	[]	[]	[]
[[2]]	[]	[]	[]
[[3]]	[]	[]	[]

4 Project Rationale

Describe the purpose, motivation, and relevance of the project. Explain why this problem matters and what you hope to learn from your research.

[Explain your motivation for choosing this research problem.]

4.1 Research Questions

State 2–4 specific, answerable research questions that your project will address. These should follow directly from the gap identified in your literature review.

RQ1. [Research question 1]

RQ2. [Research question 2]

RQ3. [Research question 3]

4.2 Aims and Objectives

List the goal of the project and the specific, measurable objectives that operationalise each research question.

O1. [Objective 1]

O2. [Objective 2]

O3. [Objective 3]

4.3 Scope of the Project

Define what will and will not be covered: specific goals, deliverables, features/functions, datasets, and any explicit exclusions. A clear scope prevents disputes at evaluation time about what was promised.

[Define the boundaries of your project – what is in scope and explicitly out of scope.]

5 Proposed Research Methodology

Explain, step by step, the systematic approach you will use to answer your research questions. Research-based FYPs should describe: research design (e.g. experimental, comparative, design-science), data/dataset, proposed model or algorithm, tools, and how you will evaluate results. Include a block diagram, flowchart, or system architecture figure.

5.1 Research Design

[Describe your overall research design/approach.]

5.2 Dataset(s)

Name the dataset(s), source/link, size, and any preprocessing you plan.

[Dataset name, source link, and description.]

5.3 Proposed Approach / System Architecture

[Describe the proposed model, algorithm, or architecture. Insert a diagram below.]

Figure 1: Proposed system architecture

5.4 Evaluation Metrics

State how you will measure success – e.g. accuracy, F1-score, precision/recall, latency, statistical significance tests, or qualitative criteria – and against what baseline(s).

[List your evaluation metrics and baseline(s) for comparison.]

6 Expected Outcomes / Research Contribution

State clearly what new knowledge, artefact, or evidence this FYP will produce, and how it advances on the related work reviewed in Section 3.

[List the expected contribution(s) of this research.]

7 Individual Tasks

Mandatory if this is a group project. List each member's tasks and justify the planned one-year effort split across FYP-I and FYP-II.

Team Member	Activity	Tentative Date
[Name 1]	**[Activity 1]**	[]
	[Activity 2]	[]
	[Activity 3]	[]
[Name 2]	**[Activity 1]**	[]
	[Activity 2]	[]

8 Timeline (Gantt Chart)

Mandatory. Show which tasks run in which period (start/end dates), which run in parallel vs. series, and the duration of each. A simple table is provided below; replace it with a proper Gantt chart (e.g. using the `pgfgantt` *package) if your department expects one.*

Table 2: Milestones for FYP-I

Duration	Milestone
[Week 1–2]	Literature review and finalisation of research questions.
[Week 3–4]	Dataset acquisition and preprocessing.
[Week 5–8]	Design of proposed methodology / model.
[Week 9–12]	Proof-of-concept implementation.
[Week 13–14]	Preliminary evaluation and results.
[Week 15–16]	FYP-I report writing and proposal defence.

9 Tools and Technologies

List the programming languages, frameworks, libraries, hardware, and any other tools the project will use.

- **[Programming language(s)]**
- **[Frameworks / libraries]**
- **[Development environment / hardware]**

10 References

Use IEEE referencing style, numbered in the order first cited in the text (Word's References tab, IEEE style, can generate this automatically if you are working in Word instead of LaTeX). Every reference listed here must be cited at least once in the body text above. Delete the two example entries below and replace them with your own.

References

[1] **[A. Author, B. Author, "Title of the paper,"** ***Journal/Conference Name*****, vol. X, no. Y, pp. XX–YY, Year.]**

[2] **[C. Author, "Title of the paper," in** ***Proc. Conference Name*****, City, Country, Year, pp. XX–YY.]**

Formatting Guidelines for This Template

These guidelines are already applied by this template's styles – keep them consistent if you add new content.

- *Font: Times New Roman throughout (applied via the `times`/`mathptmx` packages).*
- *Heading 1 (`\section`): 16pt, bold.*
- *Heading 2 (`\subsection`): 13pt, bold.*
- *Heading 3 (`\subsubsection`): 12pt, bold italic.*
- *Body text: 12pt (via `mathptmx`), single line spacing.*
- *Figure captions: below the figure. Table captions: above the table. Cross-reference every figure/table in the text (e.g. "see Fig. 1").*
- *References: IEEE numbered style, cited in order of first appearance.*
- *Margins: 1 inch on all sides, A4 paper (set in `uolfyp.cls`).*